

Caring for Children with Medical Complexity

Shreeti Kapoor, MD • Children's Hospital of Georgia • slide companion for the 30-slide session

HPC session

Not core lecture #10

3 faculty objectives

Family-centered care

Care coordination

Factoid sheet companion

Overview: Faculty Learning Objectives

Objective 1. Develop an approach to managing children with complex comorbidities.

Objective 2. Understand the unique needs of medically complex children.

Objective 3. Describe the patient- and family-centered care model.

HOW TO STUDY THIS HPC SESSION

This is **not another disease-memorization lecture**. The high-yield task is to build a usable clinical framework: recognize complexity, anticipate fragmentation, coordinate across settings, include the entire family, and make transitions/emergency plans safer.

THE WHOLE SESSION IN ONE LINE

Complexity is not just “many diagnoses.” It is the interaction of chronic conditions, functional limitations, high health-care use, technology dependence, and substantial family/service needs.

Slides 4-11

Who is a child with medical complexity?

Objectives 1-2



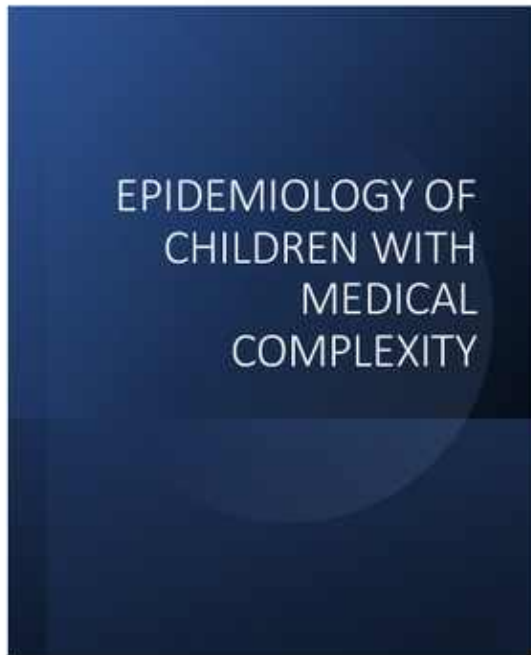
- Child with Complex Comorbidity:
- In 1998 this term was defined by the Maternal and Child Health Bureau
- “Those who have one or more chronic physical, developmental, behavioral or emotional conditions and who also require health and related services of a type or amount beyond that required by children generally”

Lecture slide 4 - full slide retained. Kapoor begins with the Maternal and Child Health Bureau framing: chronic physical, developmental, behavioral, or emotional conditions plus health-related service needs beyond those of children generally.

Slide 5: complexity can come from many different disease pathways

- Congenital or genetic anomalies.
- Significant prematurity with subsequent health complications.
- Chronic childhood medical conditions.
- Malignancy.
- Brain/spinal cord malformations.
- Cardiac and respiratory malformations.

The point is **not** to memorize a disease list. Different diagnoses can create a similar care burden and similar coordination needs.



- 1-2% of the total pediatric population
- Prevalence in Georgia similar to US data
- ACES and children with special healthcare needs
- 62.8% of all pediatric hospitalization in 2022 (up from 49% in 2009)

Lecture slide 6 - full slide retained. Kapoor frames this as a relatively small pediatric population with a disproportionately large hospitalization burden.

FACULTY NUMBERS

About 1-2% of the pediatric population in the slide, yet the deck cites **62.8% of pediatric hospitalizations in 2022**, up from 49% in 2009.

DO NOT STEREOTYPE WHAT COMPLEXITY “LOOKS LIKE”

Slides 7-8 deliberately show variation. A child may use a wheelchair, have cognitive or physical differences, depend on technology, or have none of those features that are obvious at first glance. **The chart, function, technology, family needs, and care burden matter more than appearance.**

Poll Question

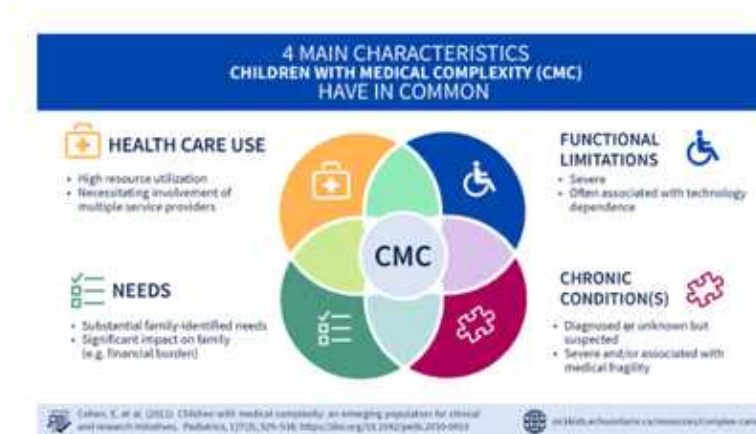
- You are in pediatric outpatient clinic and are getting ready to go see a patient. You notice that there are 2 patients scheduled with a chief complaint of "cold symptoms". You look through the chart and notice that Patient A is otherwise healthy with no significant past medical history and Patient B has medical complexity with the following attributes: carbohydrate deficient glycoprotein syndrome, end-stage renal disease, epilepsy and dilated aortic root; relies on a G-tube for feeding and has a tracheostomy for oxygenation and ventilation, and numerous chronic medications as well as residing with parents who are struggling with marital, employment and financial difficulties.
- Which patient would you go see?
- Patient A
- Patient B

Lecture slide 9 - full slide retained. The poll contrasts an otherwise healthy child with a medically complex child who has multiple chronic conditions, a G-tube, tracheostomy, many medications, and major family stressors. The same chief complaint can represent very different risk.

CLINICAL MEANING OF THE POLL

"Cold symptoms" are not automatically low acuity in a technology-dependent child. Before entering the room, you need to know the child's **baseline physiology, devices, medications, specialists, prior decompensation patterns, and caregiver concerns.**

Defining Children with Medical Complexity



Lecture slide 11 - full slide retained. The slide's visual framework centers four overlapping features: chronic conditions, functional limitations, substantial family/service needs, and high health-care use.

Slide 10: disability-based discrimination

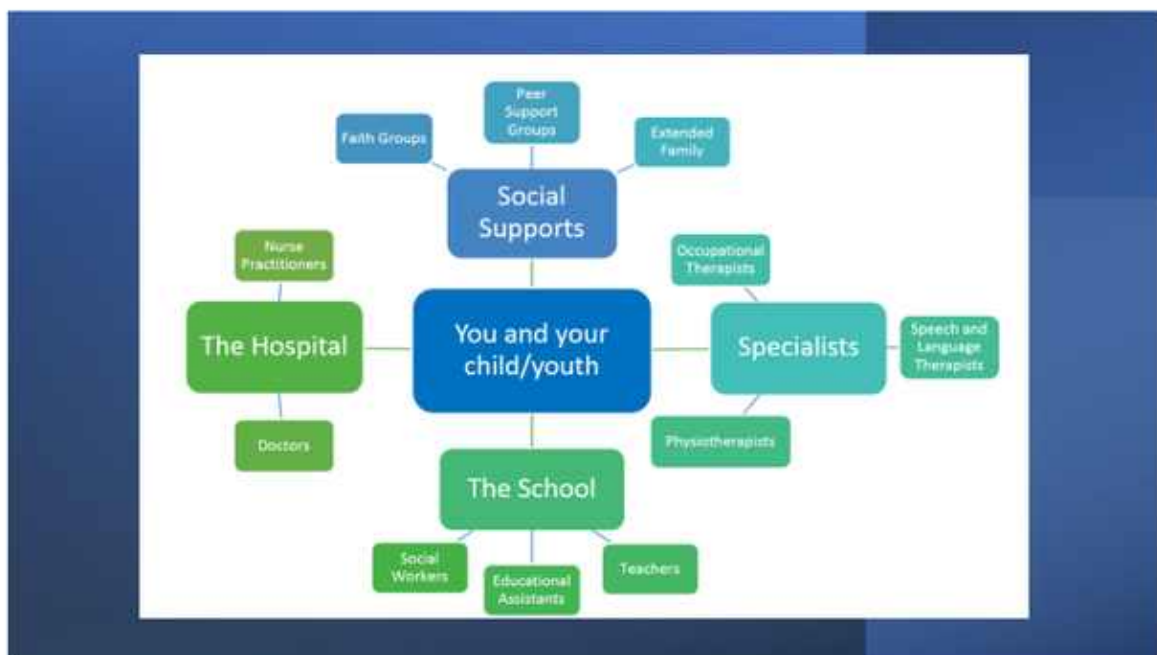
Kapoor includes a Pediatrics figure on **perceived disability-based discrimination in health care for children with medical complexity**. The slide functions as a reminder that communication, assumptions about disability, and how clinicians value quality of life can themselves become barriers to equitable care. The figure is not reproduced here because its small text is not reliably legible at study-guide scale; its core topic is preserved.

FOUR-DOMAIN DEFINITION

Chronic condition(s) + functional limitations + high health-care use + major family/service needs.

Slide 12: outpatient care is coordination-heavy

- Primary care physician.
- Tertiary-care specialists.
- Therapists.
- Technology support, including tracheostomy and enteral feeding.
- Care coordination across all of the above.



Lecture slide 13 - full slide retained. The diagram shows the child/family connected to hospital teams, specialists, school supports, and social supports. No single clinician can safely function as an isolated silo.

THE PCP'S ROLE

The primary-care clinician is often the longitudinal anchor: preventive care, ordinary childhood illness, surveillance of chronic issues, medication reconciliation, referrals, and coordination when specialty plans collide.

Slide 14: the patient is not the only person carrying the illness burden

- Caregiver stress can significantly affect mental health.
- Siblings can also experience psychosocial effects.
- The slide reports that **over 50% of families** experienced financial difficulties related to caring for a child with medical complexity.

FAMILY ASSESSMENT IS PART OF THE MEDICAL ASSESSMENT

Ask what the family can realistically carry: **time, transportation, work, finances, sleep, other children, equipment, medication access, home nursing, and caregiver confidence.**

Slide 15: collaborative improvement

The Collaborative Improvement and Innovation Network slide visually centers **family focus groups, reflection, parent voice, peer support, and care-system redesign**. The practical message is that improvement work should be built **with families**, not only designed for them.



-LISTEN
-TEACH
-CALM

Lecture slide 16 - full slide retained. Kapoor condenses the parent perspective into three words: LISTEN, TEACH, CALM.

Ozzy checkpoint - click him once you can explain why “listen, teach, calm” is a clinical skill rather than just bedside manner.

Parent message	Clinical translation
LISTEN	Parents/caregivers often know the child's baseline, device quirks, and early warning signs better than a new clinician.
TEACH	Explain the plan, equipment, medications, warning signs, and what to do when the plan fails.
CALM	Complex situations are cognitively and emotionally demanding; clear, steady communication reduces error and fear.

Emergency Care Planning for Children with Medical Complexity

- Norah's Story (Pediatrics November 2025)
- Challenges: Fragmented Care, Medication Availability, Technology Dependence
- 80% of CMC present to freestanding ED for emergency care
- Solutions: Systematic ED Care Plan, Ready-to-go Access Kit, On-site Respiratory Care Box, Care Coordination Meetings

Lecture slide 17 - full slide retained. Kapoor highlights fragmented care, medication availability, and technology dependence as emergency-care challenges, then proposes standardized ED plans, ready-to-go access kits, respiratory-care resources, and care-coordination meetings.

EMERGENCY PLAN SHOULD ANSWER

What is baseline? What devices are present? What medications are time-critical? What are the child's common emergencies? What interventions work or do not work? Who must be called?

Slides 18-20

Inpatient care: treat the child as a whole

Objectives 1-3

Slide 18: frequent admissions change the child's life

Hospitalization is not a neutral event. Repeated stays interrupt home routines, school, therapy, sibling relationships, caregiver employment, sleep, and development.

“Another admission” can be medically routine to the team and still be a major family disruption.



Lecture slide 19 - full slide retained. Kapoor's inpatient framework has three pillars: think of the child as a whole, use a multidisciplinary care team, and provide family-centered care.

Pillar	What it prevents
Whole-child thinking	Treating only the admitting diagnosis while missing nutrition, development, communication, mobility, comfort, school, or psychosocial needs.
Multidisciplinary team	Conflicting plans and fragmented specialty care.
Family-centered care	Making technically correct plans that cannot be safely implemented at home.



Lecture slide 20 - full slide retained. The slide reinforces that families need practical preparation, organization, education, professional relationships, and self-care - not just a discharge summary.

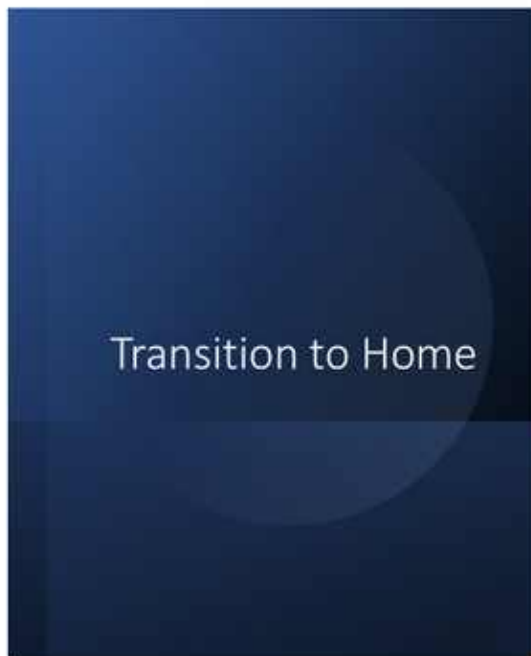
FAMILY-CENTERED DOES NOT MEAN “THE FAMILY DECIDES EVERYTHING”

It means the family is a genuine partner in information sharing, planning, implementation, and shared decision-making. The medical team still provides expertise and safety boundaries.

Slides 21-22

Transition to home and caregiver workload

Objectives 1-3



- Limited Home Nursing Care
- Varies from State to State
- Medicaid can sometimes provide better care than private insurer
- Training for caregivers essential-use of technology, administering medications

Lecture slide 21 - full slide retained. Kapoor emphasizes limited and state-variable home nursing, insurance differences, and the need for caregiver training in technology and medication administration.

DISCHARGE READINESS IS A CAPABILITY QUESTION

Not just “is the child medically stable?” Ask: **Can the caregivers safely perform the required care with the equipment, medications, nursing support, transportation, supplies, and backup plan actually available?**



- Avoiding isolation
- Organization of all the needed items for the child
- Settling into a daily routine
- Juggling the medically complex child's healthcare needs with those of other family members

Lecture slide 22 - full slide retained. The caregiver slide focuses on avoiding isolation, organizing supplies, creating routines, and balancing the medically complex child's needs with the rest of the family.

Transition failures often occur between systems

- Hospital -> home.
- Specialist -> PCP.
- Home -> school.
- Day team -> night caregiver.
- Routine care -> emergency department.

The safer strategy is to **make handoffs explicit**: medication list, device plan, supplies, contingency plan, follow-up ownership, and who is responsible for each pending issue.

Slide 23: integrate palliative care because it can improve quality of life

The lecture does **not** present palliative care as synonymous with stopping treatment or imminent death. It is a supportive discipline that can be integrated alongside disease-directed care.



- Symptom Management
- Psychosocial Support
- Communication
- Care Coordination
- Shared Decision Making
- Spiritual Support

Lecture slide 24 - full slide retained. Kapoor lists symptom management, psychosocial support, communication, care coordination, shared decision-making, and spiritual support.

PALLIATIVE CARE DOMAINS

Symptoms • psychosocial support • communication • coordination • shared decisions • spiritual support.

COMMON MISCONCEPTION

Palliative care can coexist with intensive, life-prolonging, or disease-directed treatment. In this session its purpose is to improve quality of life and support complex decision-making for the child and family.

SLIDES 25-26: REFLECTION RATHER THAN TESTABLE FACTS

Slide 25 is a memorial/personal reflection image connected to the human impact of complex pediatric illness, and Slide 26 contains an embedded media clip. I have **not reproduced the memorial image** in the study guide, and the PDF does not provide a transcript for the video, so I am not inventing content from it. Treat these slides as reflective extensions of the palliative/family-centered themes above rather than memorization targets.

Clinical synthesis

**A practical approach to the medically
complex child**

All
objectives

#	Ask	Why
1	What is this child's baseline?	Vital signs, respiratory support, neurologic status, feeding, mobility, communication, and typical behavior can all differ from population norms.
2	What technology is keeping them stable?	Trach/ventilator, oxygen, G-tube, pumps, catheters, seizure rescue meds, etc. create device-specific failure modes.
3	What medications are essential and time-sensitive?	Polypharmacy increases omission, interaction, availability, and reconciliation risks.
4	Who knows the child best?	Caregiver knowledge is a major clinical data source; specialists/PCP may also hold important longitudinal context.
5	What systems must coordinate today?	Hospital, specialists, PCP, school, therapists, home nursing, pharmacy, equipment suppliers, social supports.
6	Can the plan actually work for this family?	Transportation, work, finances, siblings, health literacy, home nursing, caregiver fatigue, and supplies determine real-world safety.
7	What is the contingency plan?	Families need explicit warning signs, emergency actions, and who/where to call.

A USEFUL MENTAL MODEL

Baseline + technology + meds + caregivers + coordination + family capacity + backup plan.

Learning-Objective Synthesis

Faculty objective	What you should be able to do without notes
Manage complex comorbidities	Start with baseline/function, technology, medication reconciliation, multidisciplinary ownership, and explicit emergency/discharge plans.
Understand unique needs	Recognize functional limitations, technology dependence, disproportionate health-care use, fragmentation risk, caregiver burden, sibling impact, and financial/social barriers.
Patient/family-centered care	Use caregivers as partners and experts on baseline, communicate clearly, teach practical skills, include the whole family, and use shared decision-making.

LAST 60 SECONDS

CMC is a care-system problem as much as a diagnosis problem.

Know the baseline. Respect caregiver expertise. Reconcile meds/devices. Coordinate teams. Plan emergencies and transitions. Check whether the home plan is feasible. Include siblings/caregivers. Use palliative care for quality of life and support - not only end-of-life care.

Source hierarchy: Kapoor's 30-slide "Caring for Children with Medical Complexity" session is the source for this companion. This is a separately labeled Holistic Patient Care resource and is not counted as a tenth core Hematology lecture.